

Education

2025–(2027) Collège Jean-de-Brébeuf – IBDP + DEC Pure and Applied Sciences (pre-uni)

2020–2025 Collège Regina Assumpta – Natural Sciences and Robotics

Research Experience

2025–present **CHUM Research Center**

- Processed lung ultrasound elastography sequences, segmented the pleural line, and ran strain analysis in MATLAB to study pleural strain's relation to tidal volume.
- Calibrated spirometry equipment and tuned ultrasound parameters for a biplanar-probe study.
- Observed in the Operating Room and Intensive Care Unit at the CHUM, including cardiac surgery (CPB, TEE), epidural placement, intubation, and continuous renal replacement therapy.

Publications/Abstracts

- Sep 2025 · CHUM Research Center *Relation entre la déformation pleurale régionale mesurée par élastographie pulmonaire et le volume courant chez des volontaires sains*. S. Chraïbi, K. El Berhoumi, A. Danatoiu, J. Fu, P. Vianna, M.H. Roy Cardinal, G. Cloutier, M. Girard (presenting author). Annual Congress of the French Society of Anesthesia and Critical Care (SFAR), Paris, France.

Work Experience

2023–present **DODO Wheels / PublicoTires – Software Developer**

- Collaborating with a small team to build a centralized operations platform for the company.
- Integrated the company website with QuickBooks Online, enabling programmatic management of invoices, customers, and inventory.
- Automated the migration of 3,000+ products and images to the new system via a web scraper.
- 2024 Developed the Doublestar Tire Canada website — doublestartyre.ca

2022–present **FIRST Robotics Competition – Mentor & former Student of Team 3990**

- Created coursework and taught programming, computer vision and AI/ML to junior team members.
- Developed computer vision object tracking and autonomous behaviors on the robot.
- Multiple consecutive Regional Winners; 2024 World Championship Impact Award finalist.

Volunteer Experience

- 2025–(2027) Mentoring the new CRC Robotics team at Brébeuf. Rookie Team of the Year award.
- 2026 Built an email-to-publish system for Collège Bois-de-Boulogne's student journal, *L'infomane*.
- 2026 Played music with a student volunteer group for patients and families at CHU Sainte-Justine.
- 2025 Advised on the development of Collège Jean-de-Brébeuf's student journal, *Le Graffiti*.
- 2025 Contributed to the organization of Team 3990's in-house off-season robotics competitions.
- 2025 Guided incoming students as a peer mentor (*Anges gardiens*) at Collège Regina Assumpta.

Extracurriculars/Notable Projects

- 2026 **WRCC Brain–Computer Interface Competition (by Tsinghua University) – Grand Prize winner**. Built an EEG motor-imagery classifier using a Riemannian geometry pipeline (4 channels), averaging 77% accuracy (48.6% baseline) across four binary tasks.

- 2025–present **Medical Inference via Vector Embeddings** – An AI system that generates a differential diagnosis from patient-reported symptoms, fusing a biomedical language model with UMLS medical concepts. Explores whether embedding geometry can align symptom combinations to the correct diagnosis. IB Extended Essay in Computer Science.
- 2026 **HiddenGarden** – Built the storefront for a local florist business, supporting Shopify's Storefront API for checkout — hiddengarden.pages.dev
- 2025–2026 **ORCA – Operational Intelligence Platform** – Designed a distributed pipeline ingesting telemetry, sensor, and radio data through an entity graph to produce real-time, cited assessments. Prepared a TRL-3 proposal for the Canadian DND/CAF IDEaS program.
- 2024 & 2025 **Secondaire en spectacle** – Performed piano-comedy skits with a friend, 400+ audience. Local winner.

Software Skills

- **Languages:** TypeScript/JS, Python, Java, Kotlin, Swift, MATLAB
- **Libraries:** React, Express.js, PyTorch, OpenCV, WPILib, Kotlin Multiplatform, SwiftUI
- **Infrastructure:** Docker, Cloudflare, GCP, OAuth, QuickBooks Online

Other Skills

- Machining and fabrication, 3D printing, laser cutting, CNC, lathe; soldering and crimping
- Basic CAD (Onshape, Autodesk Inventor)
- Photography and Adobe Lightroom

Certifications

- **Research ethics:** Good Clinical Practice (CITI), Éthique de la recherche (MSSS), Responsible Conduct of Research (CRCHUM)
- **Canadian Red Cross:** Standard First Aid & CPR/AED Level C, Psychological First Aid